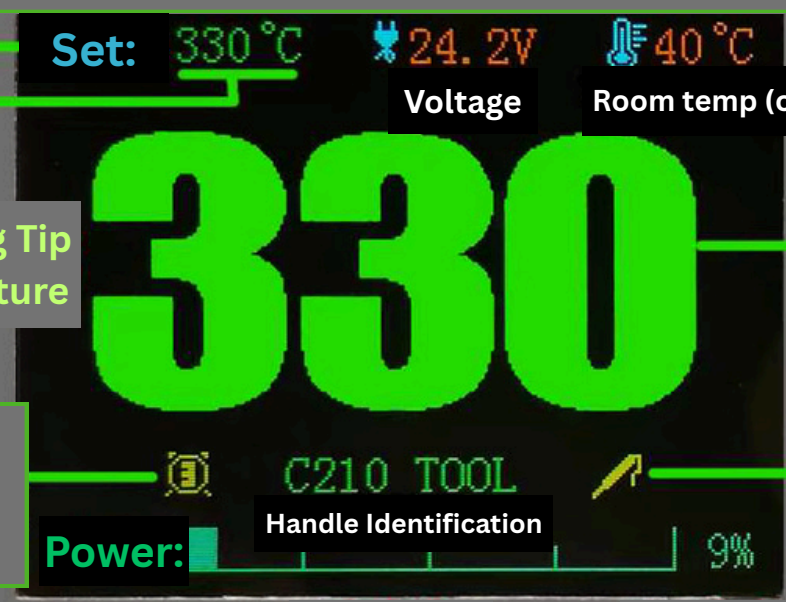
Short-press the encoder or the CONFIRM button to toggle between **Set Temperature mode** and **Preset Temperature mode. Set Temperature mode —** rotate the encoder left/right to adjust the temperature. The step value is set in *Temperature Settings → Temperature Step*. **Preset Temperature mode —** configure presets in *Main Menu → Preset Temperature → Preset 1–4* (set your commonly used temperature combinations).



Current Set Temperature

Voltage

Room temp (cold-junction compensation)

Soldering Tip Temperature

Temperature color ranges:
0–180 (cool),
180–320 (warm),
320–400 (hot),
400–500 (high-temp, warning),
500+ (danger).

Iron Sleep Countdown Icon

Handle Identification

This icon indicates vibration-sleep mode and shows the on/off status of the mercury/vibration switch.

Power bar

Real-time power

Project Link

Sleep interface

Sleep State temperature

Remaining sleep time

Soldering Tip Temperature

Default is vibration-based sleep mode. Default detection time is 3 minutes. If no handle vibration is detected within 3 minutes, the system automatically enters in sleep state.

Move the handle or operate the rotary encoder to exit sleep state.

Handle not connected

Error Message

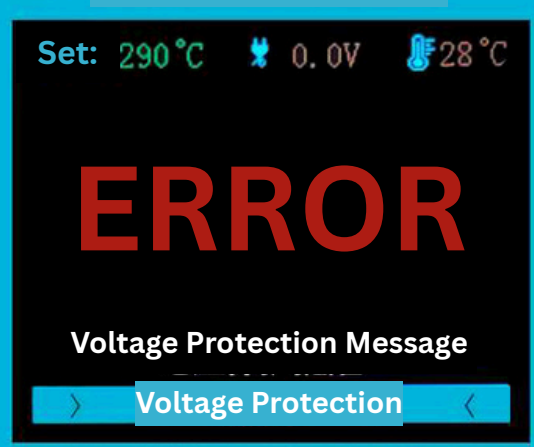
This error message will only appear when the handle is not connected or if there is a problem with the handle connection.

Manual sleep mode: *Main Menu → Time Settings → Sleep Time* = set to Manual.

Vibration sleep mode: *Main Menu → Time Settings → Sleep Time* = set to 1–99 (default 3).

"1–99" is the vibration detection duration (in minutes); the default is 3 minutes.

Voltage Protection



Voltage Protection Message

The voltage value at which the soldering iron stops heating. When the supply (or battery) voltage falls below this set value, the iron will stop heating, but this does not affect menu operations.

Related setting: *Main Menu → System Settings → Protection Voltage*

Other Functions Introduction:

Encoder/button calibration: The A/B channels of encoders from different manufacturers may not be identical. If the encoder direction does not match your preference, change this option to 1 (default is 0).

Sleep temperature: The working temperature of the iron when it is in Sleep Mode. If this option is set to less than 50°C, the iron will not heat during sleep.

Tip selection: Allows the user to choose calibration data for T12 / JBC245 / JBC210 / 936 tips, and users can also recalibrate their own tips.

Maximum temperature: The highest temperature that can be set for the iron; this option must not exceed 500°C, accuracy decreases above 400°C, and prolonged high-temperature operation will reduce tip life, please use with caution!

Tip Temperature Calibration Procedure:

Enter Tip Calibration; then choose Tip Select/Cal, rotate the encoder or use the buttons to select the tip model you want to calibrate, and click the encoder or press CONFIRM to save!

Next, click to select the temperature options from 100°C to 500°C one by one — the iron will heat according to the temperature table and will automatically stop when an option is deselected.

During calibration, if the measured temperature is lower than the temperature in the table you selected, increase the AD value; if the measured temperature is higher, decrease the AD value.

Note: A dedicated iron thermometer is required/should be used during the temperature calibration process.

Basic Operations Guide:

- On the main interface, single-click the encoder/OK button to activate the one-key temperature increment.
- On the main interface, double click the encoder/OK button to activate the preset temperature mode.
- On the main interface, long press the encoder/OK button to enter the system menu.
- On the main interface, rotate the encoder or briefly press the increase/decrease button to adjust the tip temperature.

Settings Menu Navigation Guide:

- Long-press the encoder or quickly press the confirm button to enter the settings menu.
- Rotate the encoder or press the increase/decrease buttons to select an item.
- Short press the encoder or press the confirm button to select an item for setup.
- After completing, short-press the encoder or the confirm button again to save.
- Long-press the encoder or the confirm button to return to the previous level.

System Control Mode Setting:

- Navigate to *Settings → System Settings → Control Mode*
- This setting let the user to enable or disable ID recognition functionality. It has two modes: Automatic and Manual.
- **Automatic mode:** Handle ID recognition control.
- **Manual mode:** Select the heater cartridge type manually.

Note: After selecting manual mode, the heater cartridge type must be manually chosen.

Additional Functionality (Available on XPG v4 Controller and Above):

Auto Power Off

- On the XPG v4 controller, pin “A” serves as the System Power-off Signal Output.
- Pin “A” is HIGH when the system is ON and LOW when it is OFF.
- An active-high relay can be connected to this pin to control the power supply unit’s ON/OFF state.
- Figure 1 illustrates a schematic demonstrating how to implement this functionality.

Auto Voltage Switching

- On the XPG v4 controller, pin “B” acts as the Auto Voltage Switching Signal Output.
- Pin “B” outputs a LOW signal when the T210 handle is connected and a HIGH signal when the T245/T12 handle is connected.
- An active-high relay can be connected to this pin to toggle between 12V and 24V power sources.
- Figure 2 illustrates a schematic for auto voltage switching using two separate power sources (12V and 24V).
- Figure 3 illustrates a schematic for auto voltage switching when a single 24V power source and a DC-DC buck converter module are used.

If the Sleep Time is set to Manual and the sleep detection input pin is at a low level— Let’s say because the soldering iron is placed in the cradle or if its tip touches the cradle’s tip change area—then the soldering station will immediately enter sleep mode.

Project Link